

Adaptations to diving in the harbor seal: oxygen stores and supply

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PACKER, BERNARD S., MICHAEL ALTMAN, CARROLL E. CROSS, H. VICTOR MURDAUGH, JR., JAMES M. LINTA, AND EUGENE D. ROBIN. *Adaptations to diving in the harbor seal: oxygen stores and supply.* Am. J. Physiol. 217(3): 903-906. 1969.—Total-body exchangeable O₂ stores and diving versus nondiving circulating blood volume were measured in the harbor seal *Phoca vitulina*. The total body exchangeable O₂ stores were 0.79 liter (40 ml/kg), (N = 3). The lung stores were 0.26 liter (14 ml/kg) and the nonlung stores were 0.53 liter (26 ml/kg). These values are approximately double those found in the dog and in man. Measurement of circulating blood volume during and after diving revealed only small changes demonstrating that the entire blood volume of approximately 3 liters is available to the animal during the dive. Oxygen consumption during diving is estimated at 50 ml/min and maximal diving time is approximately 20 min. This period corresponds to the time required for exhaustion of total-body O₂ stores.

blood volume; aquatic mammals

THE ADAPTATIONS that permit survival during prolonged cessation of external O₂ supply in the diving mammal have been studied extensively (1, 15). These adaptations serve to continue the perfusion of the central nervous system, lungs, and probably heart with virtual cessation of blood flow to all other organs (3, 5, 9). In the present study, two previously unanswered questions have been approached: 1) What is the magnitude of the total exchangeable O₂ stores? 2) What fraction of these stores is available for O₂ metabolism in critical areas during diving? The data indicate that the maximal length of diving corresponds to the time when available body O₂ stores are consumed.

METHODS AND MATERIAL

All studies were performed on the harbor seal, *Phoca vitulina*. Exchangeable O₂ stores were measured in three animals by a method based on the dilution of the stable oxygen isotope, ¹⁸O₂, under circumstances in which ¹⁸O₂ is introduced in tracer amounts through a rebreathing circuit (4). A closed rebreathing circuit consisting of a face mask, a CO₂ absorber, and a rebreathing bag was used. The face mask was designed to fit tightly over the animal's head and a water seal assured no leakage. The rebreathing bag was initially filled from a spirometer with ¹⁶O₂ (60-90%),

¹⁸O₂ (0.5-1.5%), neon (1-2%), and the balance nitrogen. The large fractional ¹⁶O₂ concentrations in the external circuit were used so that studies could be continued for as long as 20 min. The ¹⁸O₂ was obtained as 93% atom ¹⁸O (86% mass 36) from Yeda Research and Development, Weizmann Institute, Rehovoth, Israel.

After an adjustment period of breathing room air through the face mask, the animal was connected to the rebreathing system. Samples were drawn periodically from the rebreathing bag into evacuated tonometers and subsequently analyzed with a model 21-103C Consolidated Electrodynamic mass spectrometer. Masses in the range 20-44 were scanned. From the relative intensity of ions mass 36 and mass 32, the tracer ¹⁸O₂ isotopic ratios were calculated (as peak height 36/32). Peak heights representing intensities of ions of masses 20, 28, 40, and 44 represented Ne, N₂, Ar, and CO₂, respectively. All gas samples were collected and analyzed in duplicate, and a reproducibility in the overall analysis of 0.5% was found. Lung volumes were calculated by using a double-dilution technique, using the initial external-circuit neon and initial lung-argon or -nitrogen dilutions and solving for initial and final lung volumes (4).

Diving blood volume measurements were performed in four animals. The animals were trained to dive for periods up to 10 min on a tilt board (11). Under procaine anesthesia, a polyethylene catheter was placed in a fore flipper vein and a similar catheter in the extradural vein. Approximately 20 ml of autogenous blood was incubated for 30 min at 37°C with ⁵¹Cr (2). The ⁵¹Cr-tagged red blood cells were then injected into the flipper vein and the animal was dived immediately after the injection. Electrocardiographic monitoring provided evidence of a sustained diving bradycardia. Serial blood samples were drawn from the extradural venous catheter at 1-min intervals during the dive. All dives were terminated at 10 min. Additional blood samples were drawn at periodic intervals for 1 hr following the dive. Blood volumes were calculated by the dilution of the injected ⁵¹Cr-labeled red blood cells (2).

The effective circulating blood volume during diving could thus be compared with the blood volumes determined after the dive. Major trapping of blood during the dive would be shown by a sharp decrease in radioactivity as sequestered noncirculated blood entered the vascular system.

RESULTS

Oxygen stores. The method used in these studies for the calculation of total-body exchangeable O₂ stores requires measurement of an external circuit O₂ volume and its fractional ¹⁸O₂/¹⁶O₂ concentrations before and after equilibrium has been reached. In previous studies in man and dog, we found that the isotopic O₂ ratios in the external rebreathing circuit did not change significantly after rebreathing periods of 5 min (4). Representative data from one of the seal studies are shown in Fig. 1. Isotopic O₂ ratios are plotted against time. The achievement of virtual equilibrium at 5 min is shown. This was consistent in all animals studies. The 5-min external rebreathing circuit dilutions were chosen to represent the equilibrium isotopic O₂ concentrations in the external circuit and used for the calculation of exchangeable O₂ stores.

Table 1 summarizes the results of the lung-volume and O₂-stores measurements. Initial lung volumes averaged 1.93 liters. Since there was no way to determine the precise portion of the respiratory cycle at which these lung volumes were measured, it is not possible to ascribe an exact meaning to these volumes for comparative purposes. It seems reasonable to assume that these volumes were either equal to or greater than functional residual capacity. If so, relative lung volumes in the seal (ml/kg) are substantially greater than in man. This finding is in accord with the rather voluminous lungs of seals as seen by visual inspection at postmortem examination.

The total exchangeable O₂ stores were calculated by the assumption of instantaneous dilution (4). The lung O₂ stores were calculated by multiplying the calculated initial lung volume by 0.15. (The initial lung O₂ store is the prod-

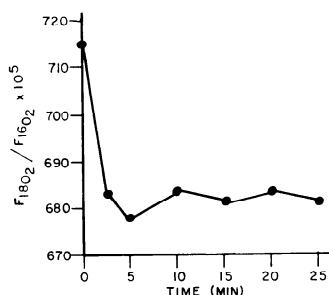


FIG. 1. Typical time-concentration curve of ¹⁸O₂ dilution in external circuit obtained in measurement of exchangeable O₂ stores in a study performed in seal 2.

TABLE 1. Lung volume and exchangeable O₂ stores in the seal *P. vitulina*

	Weight, kg	Initial Lung Volume, liters (RTPS)	Total Exchangeable O ₂ Stores		Nonlung Stores	
			liters (STPD)	ml/kg	liters (STPD)	ml/kg
Seal 1 (4)	19.2	1.73	0.75	39	0.50	26
Seal 2 (7)	20.0	1.53	0.68	34	0.49	24
Seal 3 (1)	21.9	2.53	0.93	46	0.59	28
Mean	20.4	1.93	0.79	40	0.53	26

Number of studies is given in parentheses.

TABLE 2. Exchangeable O₂ stores (ml/kg) in the seal *P. vitulina* as compared to dog and to man (4)

Seal, Hct = 60%		Dog, Hct = 45%		Man, Hct = 45%	
Total	Nonlung	Total	Nonlung	Total	Nonlung
40	26	17	10	14.7	10.9

TABLE 3. Blood volume determinations during diving and following diving in the seal *P. vitulina*

Seal	Weight, kg	Blood Volumes		
		Dive, liters	Postdive, liters	Postdive, ml/kg
1	23.6	2.55	2.94	125
2	21.8	2.90	3.02	139
3	23.6	2.98	3.00	127
4	23.6	2.64	2.81	119

The dive blood volumes were calculated on the basis of samples obtained after 10 min of diving and the postdive blood volumes were calculated on the basis of the samples obtained 1 hr after the dive was terminated.

uct of the initial lung volume and the mean lung O₂ fractional gas concentration. Since we did not determine the mean O₂ of the lung gas, the fractional concentration of 0.15 was used (12).) The coefficient of variation in the total O₂-store measurements of the seven studies performed in seal 2 was 19%. The mean total-body exchangeable O₂ for the three seals was 0.79 liters or 40 ml/kg. The mean nonlung stores were 0.53 liters or 26 ml/kg.

Table 2 presents data comparing total exchangeable O₂ stores in the seal with previous determinations by an identical method in dog and man (4). It can be seen that nonlung O₂ stores in dog (10 ml/kg) and in normal man (10.9 ml/kg) are nearly identical, whereas in seal (26 ml/kg) the nonlung stores are more than double these values.

Blood volumes during diving. Blood volumes in four seals during and following diving are shown in Table 3. During the dive, blood volumes measured 2.55 to 2.98 liters. Following the dive, the total blood volumes measured 2.81 to 3.02 liters. Although the diving blood volumes were slightly lower than the postdive volumes, the changes in the two can probably be accounted for by the increased time of equilibration for ⁵¹Cr-tagged red blood cells throughout the circulation during the 1-hr postdive period. The absolute values of the blood volume are in agreement with the estimate made by others (7). It seems clear that at least 95% of the circulating blood volume of the animal is maintained in the central circulation during the dive and that its O₂ content is available for O₂-consuming processes.

DISCUSSION

Oxygen metabolism in an intact organism may be considered to involve three variables: 1) O₂ uptake, 2) O₂ stores, and 3) O₂ utilization. In the usual steady state, O₂ uptake equals O₂ utilization and the stores remain constant. If uptake becomes zero (as in the dive), utilization

proceeds at the expense of the stores. The maximal time of diving compatible with survival depends on the magnitude of available O₂ stores and on the rate of O₂ utilization. Previous studies from a number of laboratories have shown that overall O₂ utilization during diving is decreased by the arterial constrictor response (14, 15). Measurements of red cell mass have suggested increased O₂ stores in diving mammals as compared to other mammals (7). However, no previous quantitative experimental estimation of total O₂ stores is available. There are no data on the magnitude of trapping of blood in organs whose arterial supply is intensely constricted.

Total oxygen utilization in the nondiving state in seals of this age and weight amounts to approximately 200 ml/min. This is balanced by pulmonary O₂ intake of 200 ml/min so that stores remain unchanged (13). During a dive, with the development of arterial constriction, oxidative metabolism ceases in all organs except the central nervous system, heart, and lung. Residual O₂ utilization during the dive amounts to approximately 50 ml/min. This value was determined by measuring changes in aortic or mixed venous oxygen contents as a function of time before and during the dive. Since there is no external O₂ supply during this time, and assuming a blood volume of 3 liters, diving O₂ consumption may be calculated as $[\Delta\text{O}_2 \text{ content}]_{\text{a or } \bar{v}} \times 3 \text{ liters/diving time (min)}$ (unpublished observations). External O₂ uptake is zero and oxidative metabolism must occur at the expense of O₂ stores. Since perfusion of the lung continues during diving, the O₂ in lung air is available for this purpose. The precise magnitude of this segment of O₂ stores during diving is not known. The values found in the present study were determined in the nondiving state and presumably reflect lung O₂ stores with the lung volume at functional residual capacity or somewhat greater (during inspiration). Prior to a dive these animals generally make an expiratory effort decreasing lung volume and thus decreasing available lung O₂ stores somewhat. It can be calculated that the effect of this on total O₂ stores would be quite small. Each 25% reduction of lung volume related to predive expiration would reduce total O₂ stores by approximately 9%.

The major segment of nonlung O₂ stores is composed of dissolved and chemically combined O₂ in the blood. The close agreement between diving and postdiving blood volumes demonstrates that essentially all of the O₂ in this segment of stores is available during diving. Previous attempts to estimate the magnitude of these stores in man (12) or seal (15) have required a division of blood O₂ stores into two compartments, one compartment with an O₂ saturation of arterial blood and the other containing blood with the O₂ saturation of mixed venous blood. It is impossible to fractionate accurately how much of the total blood volume is present in each compartment. The calculation of total stores is quite sensitive to the figures arbitrarily assigned to this fractionation. In addition, estimates of body O₂ stores have, in general, included estimates of the O₂

stores in myoglobin. However, because of the high affinity of myoglobin for O₂, the precise quantity of O₂ stores to ascribe to this compartment is difficult to ascertain. The situation is even more complex in the seal since the O₂ combined with myoglobin is no longer generally available during diving because of the arterial constrictor response. The actual amount of available O₂ stores during diving would be somewhat less than the stores measured in this study but even with the high myoglobin concentration found in this animal (15), the difference should be small. The present estimate of nonlung O₂ stores is, to our knowledge, the first experimental determination of this parameter. It does seem clear that, since maximal diving time is approximately fivefold that found in man and O₂ stores are increased only 2.5 times, an increase in stores cannot account quantitatively for the increased diving time in *Phoca vitulina*.

Having considered the relationship between O₂ utilization and O₂ stores during the time of zero O₂ uptake (diving), it is useful to consider this relationship in terms of maximal diving time. In seals that fail to develop an arterial constrictor response, or in which the response has been aborted pharmacologically, death occurs within approximately 4 min. Assuming an O₂ consumption of approximately .20 liter/min and O₂ stores of approximately 0.80 liter, this would correspond to the time when O₂ stores would be entirely exhausted.

During a dive with an intact arterial constrictor response, the lower limits of residual O₂ consumption will be dictated by the magnitude of central nervous system, heart, and lung O₂ consumption. Since O₂ consumption during diving is approximately 50 ml/min and O₂ stores are approximately 0.80 liter, it would be anticipated that maximal diving time would be approximately 16 min and indeed peak diving times of approximately 20 minutes have been found experimentally (15). The absolute upper limit of diving time is then dictated by the magnitude of O₂ stores. The present data provide a quantitative basis for defining the relationship between augmented O₂ stores and the arterial constrictor response as adaptive mechanisms for prolonged diving in this aquatic species.

The precise situation in other aquatic mammals such as the Weddell seal and whales, where diving times close to an hour or longer have been recorded, is not clear (6, 10). A priori, this suggests the presence of even greater O₂ stores in these animals than in *Phoca vitulina*. Measurements of O₂ stores in these species obviously would be of interest.

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